

Software Requirements Specification (SRS)

Stock Brokerage and Portfolio Management System

1. Introduction

1.1 Purpose

The purpose of this SRS document is to define the comprehensive requirements for the Stock Brokerage and Portfolio Management System. This system will facilitate a streamlined, secure, and efficient process for managing client accounts, executing equity and derivative trade orders, tracking investment portfolios, processing settlements, and generating SEBI-compliant regulatory reports for brokerage operations.

1.2 Scope

The system enables stockbrokers, dealers, and investment advisors to manage client onboarding with KYC, execute trade orders on NSE and BSE, track portfolio performance, manage client margins, process trade settlements, and generate SEBI-mandated regulatory reports. Key functionalities include:

- Client and broker authentication with RBAC and trading permissions
- Complete CRUD for client accounts, trade orders, and portfolio holdings
- Multi-type order management: market, limit, stop-loss, and bracket orders
- Real-time portfolio valuation with MTM P&L computation
- Margin management with automatic margin call alerts and position square-off
- Trade settlement T+1 processing and brokerage ledger management
- SEBI regulatory file generation and exchange report submission
- Tax P&L statement with LTCG and STCG classification and computation
- Options and futures position management with span margin calculation
- Analytics for portfolio performance, risk metrics, and compliance monitoring

1.3 Definitions, Acronyms, and Abbreviations

- SEBI: Securities and Exchange Board of India
- NSE: National Stock Exchange of India
- BSE: Bombay Stock Exchange
- DEMAT: Dematerialised securities account
- P&L: Profit and Loss
- LTCG: Long-Term Capital Gains
- STCG: Short-Term Capital Gains
- MTM: Mark-to-Market portfolio valuation
- DP: Depository Participant
- CDSL: Central Depository Services Limited
- NSDL: National Securities Depository Limited
- XIRR: Extended Internal Rate of Return

1.4 References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications
- Spring Boot Security Documentation
- React.js Documentation
- SEBI Stockbroker Regulations
- NSE and BSE Exchange Trading Guidelines

1.5 Overview

Covers all requirements for the Stock Brokerage and Portfolio Management System with role-based security and comprehensive stock brokerage and portfolio management.

2. Overall Description

2.1 Product Perspective

Spring Boot backend + React.js frontend + JWT authentication + JPA database operations for stock brokerage and portfolio.

2.2 Product Functions

- Client onboarding with DEMAT, PAN, and Aadhaar verification
- Multi-type order management with exchange gateway integration
- Real-time portfolio MTM valuation with unrealised and realised P&L
- Margin management with span calculation and automatic margin call alerts
- T+1 settlement processing and brokerage charge ledger
- Capital gains tax P&L with LTCG and STCG computation
- SEBI daily activity report and UCC file generation
- Research analyst recommendations and model portfolio management
- Risk metrics: VaR, beta, Sharpe ratio, and portfolio concentration
- Corporate action processing: dividends, stock splits, and bonus shares

2.3 User Classes and Characteristics

- System Administrator: Full access, system configuration, and compliance oversight
- Client: Own account trading, portfolio viewing, statements, and research access
- Dealer/Broker: Client order execution, account management, and margin oversight
- Research Analyst: Research report creation and investment recommendation publishing
- Compliance Officer: Regulatory report generation and audit trail review
- Risk Manager: Margin monitoring, position limit configuration, and risk alerts

2.4 Operating Environment

- Server: Spring Boot 3.x with Spring Security (Java 17+)
- Database: JPA-compatible relational database with stock brokerage and portfolio management schema
- Frontend: React.js with authentication components (port 8081)
- Backend: Spring Boot with Spring Security REST APIs (port 8080)
- Authentication: JWT-based with role-specific token claims and expiry
- Testing: JUnit 5 for backend, React Testing Library for frontend
- Deployment: Docker containerised on cloud (AWS, Azure, or GCP)

2.5 Design and Implementation Constraints

- Spring Security + JWT on all trading and financial data endpoints
- Name: alphabetic only; Phone: exactly 10 digits
- Order execution requires sufficient margin validation before order acceptance
- PAN and DEMAT verification mandatory before trading account activation
- All executed trade orders and modifications are immutable after confirmation
- SEBI regulatory files must be generated in precisely prescribed exchange formats

2.6 User Documentation

- Client account opening and online trading guide
- Dealer order execution and client portfolio management guide
- Compliance officer SEBI regulatory reporting and audit guide

- Risk manager margin monitoring and position limit configuration guide

2.7 Assumptions and Dependencies

- NSE and BSE market data feed for real-time price and order book data
- CDSL and NSDL DEMAT holding verification API for client holdings
- PAN verification API for KYC compliance and identity confirmation
- NEFT and UPI payment gateway for client fund transfer and payout
- SEBI regulatory file upload portal for daily and periodic submissions

3. Specific Requirements

3.1 External Interfaces

3.1.1 User Interfaces

- Landing Page: Brokerage overview with client login and market summary
- Trading Terminal: Order entry, real-time order book, and executed trade log
- Portfolio Dashboard: Holdings, MTM P&L, sector allocation, and performance chart
- Research Portal: Analyst recommendations, model portfolios, and target prices
- Client Account: Fund balance, margin utilisation, and ledger statement
- Compliance Panel: Regulatory reports, audit trail, and SEBI file status
- Admin Panel: User management, system configuration, and risk parameters

3.1.2 Hardware Interfaces

- Document scanner for physical document digitisation
- Digital signature pad for contract and form signing
- Barcode/QR scanner for item and record identification

3.1.3 Software Interfaces

- Secured JWT REST API for all protected endpoints
- Email service (SMTP/SendGrid) for notifications and alerts
- SMS gateway for critical alerts and OTP delivery
- Document storage service for file upload, versioning, retrieval

3.1.4 Communication Interfaces

- HTTPS with TLS 1.3 for all client-server communications
- JSON with security headers (CORS, CSP, X-Frame-Options) on all responses
- JWT HMAC-SHA256 signature verification on every API request
- WebSocket for real-time notifications and live dashboard updates

3.2 Functional Requirements

User Authentication and Account Management

FR1: Comprehensive User Registration and Authentication System

Description: The system shall provide stock brokerage and portfolio-specific registration capturing credentials and role-appropriate information to establish trusted access.

Detailed Requirements:

- Registration form captures full name, employee or user ID, email, role type, stock brokerage and portfolio assignment, and contact details
- User verification requires email confirmation and ID validation against stock brokerage and portfolio directory

- Account activation includes automated OTP verification followed by supervisor approval for all operational roles
- Multi-role support handles System Administrator, Risk Manager, Compliance Officer, Dealer, Research Analyst, Client accounts with role-appropriate verification
- Password enforces minimum 8 characters with uppercase, lowercase, digit, and special character
- Account recovery uses time-limited email OTP with 15-minute validity window
- Profile management allows post-registration updates with full audit trail
- Security logging records every authentication event with IP address and timestamp
- Two-factor authentication optional for standard roles; mandatory for financial and admin operations
- Multi-role users can switch context between roles within a single authenticated session

FR2: Secure Authentication and Session Management System

Description: The system shall implement robust authentication for stock brokerage and portfolio operations maintaining high security standards.

Detailed Requirements:

- JWT tokens configured with role-specific expiry durations for appropriate session length
- Multi-credential login accepts user ID, email address, or registered mobile number
- Session enforces automatic logout after 30 minutes of user inactivity
- Device recognition tracks registered devices and requires extra verification for new access points
- Progressive logout: 15, 30, and 1440 minutes after 5, 10, and 15 consecutive failed attempts
- Single sign-on integration supports institutional or corporate authentication systems
- Security logging records every login, logout, session timeout, and forced logout event
- Account lockout prevents brute force attacks with emergency override for critical scenarios
- Token blacklisting implemented on logout, password change, and security breach events
- All session activities in stock brokerage and portfolio operations logged for compliance and audit trail

FR3: Role-Based Access Control System

Description: The system shall implement granular RBAC ensuring users access only data within their assigned scope.

Detailed Requirements:

- Client role: own holdings, P&L, orders, and statements; Dealer: assigned client order execution and portfolio management; Risk Manager: all client margin positions and risk monitoring
- Hierarchical permission structure supports configurable permissions per role and function
- Temporal access controls enforce operational hours and emergency access override procedures
- Function-based permissions control access to specific sensitive and financial operations
- Data access controls ensure users see only data relevant to their role and assignment
- Audit trail logs every data access event with user identity, timestamp, and declared purpose
- Role changes take effect on the user's next login session after administrator update
- Cross-department data access requires documented justification and management approval
- Guest access restricted to public information with no access to operational data
- Permission matrix maintained in database for runtime enforcement without code changes

Trading, Portfolio, and Settlement Management

FR4: Order Management and Trade Execution System

Description: The system shall provide a complete order management system supporting all order types with real-time exchange integration.

Detailed Requirements:

- Order entry supports market, limit, stop-loss, bracket, and cover order types with full parameter validation
- Margin validation at order placement checks available margin against the computed order value in real time
- Real-time order book displays all pending, executed, modified, and cancelled orders per client
- Modify pending order allows price and quantity changes before exchange acknowledgement of execution
- Exchange gateway integration submits and receives order confirmations from NSE and BSE via FIX protocol
- Order execution confirmation updates holdings, ledger, and portfolio MTM P&L instantly on receipt
- Circuit breaker detection rejects new orders for stocks currently in upper or lower circuit halt
- GTT (Good Till Triggered) orders persist across trading sessions until triggered or manually cancelled
- Order audit trail records every placement, modification, cancellation, and execution with timestamp
- Bulk order import from CSV file supports algorithmic and basket trading strategies for dealers

FR5: Portfolio Valuation and Performance Analytics System

Description: The system shall provide real-time portfolio MTM valuation with comprehensive investment performance analytics.

Detailed Requirements:

- MTM valuation refreshes holding values every 15 seconds during active market trading hours
- Unrealised P&L computed per holding: (current_price - avg_cost) multiplied by quantity held
- Realised P&L accumulated from all closed positions since the start of the current financial year
- Portfolio return metrics: absolute return, CAGR, and XIRR calculated from inception date
- Sector and market capitalisation allocation pie charts for portfolio diversification analysis
- Holdings performance comparison against selected benchmark index (Nifty 50 or BSE Sensex)
- Risk metrics: portfolio beta, Sharpe ratio, Sortino ratio, and Value-at-Risk at 95% confidence level
- Historical performance chart available for 1W, 1M, 3M, 6M, 1Y, and custom date range selection
- Top 10 portfolio gainers and top 10 losers sorted by absolute and percentage gain or loss
- Asset class breakdown: equity, mutual funds, bonds, gold, and cash allocation summary view

FR6: Margin Management and Risk Control System

Description: The system shall manage client margins in real time with automated risk alerts and position square-off controls.

Detailed Requirements:

- Available margin calculated continuously: ledger balance plus approved collateral minus utilised margin
- Span margin and exposure margin computed for F&O positions per SEBI VaR norms daily
- Real-time margin utilisation percentage displayed prominently on the trading terminal header
- Margin call notification at 80% utilisation dispatched via SMS and push to client immediately
- Automatic position square-off initiated at 90% utilisation if client has not topped up margin
- Collateral pledge workflow: client pledges SEBI-approved securities for additional trading margin
- Mark-to-market loss computation for overnight F&O positions with daily settlement to ledger
- Margin shortfall report generated daily for Compliance Officer and Risk Manager review
- Risk parameter configuration panel for VaR percentage, exposure limits, and concentration limits
- Client-level position limit enforcement per SEBI large trader threshold reporting norms

FR7: Trade Settlement and Brokerage Ledger System

Description: The system shall process T+1 trade settlements and maintain accurate client ledger accounts.

Detailed Requirements:

- T+1 settlement: sale proceeds and purchase debits reflected in client ledger the next trading day
- Brokerage charge computation per configured rate: flat fee per order or percentage of turnover
- GST at 18% computed on brokerage with itemised GST invoice generated for every settlement
- Exchange transaction charges, STT, stamp duty, and SEBI turnover fees automatically computed
- Fund transfer: NEFT and UPI integration for client fund credits and payout request processing
- Corporate action processing: bonus shares, stock splits, dividend credits, and rights issue management
- Daily ledger reconciliation: automated matching of all debits and credits to detect discrepancies
- Client account statement generated in PDF and CSV for daily, monthly, or custom date ranges
- Payout instruction processing with bank account verification via penny drop before NEFT execution
- IFRS-compliant portfolio accounting with accurate gain and loss classification per holding

Compliance, Research, and Client Services

FR8: SEBI Regulatory Reporting and Compliance System

Description: The system shall generate all SEBI-mandated regulatory reports and exchange submission files.

Detailed Requirements:

- Daily Activity Report generated in exchange-prescribed format and submitted by end of every trading day
- Unique Client Code file submitted to exchange automatically on activation of each new client account
- CKYC (Central KYC) upload for client KYC records to the national CKYC registry
- PMS reporting for discretionary and non-discretionary portfolio management clients
- Suspicious Transaction Report flagging workflow for AML compliance officer review
- Annual account statement mandatory generation and distribution to all active client accounts
- MTF (Margin Trading Facility) utilisation report per SEBI Margin Trading guidelines
- Client fund segregation audit: verification that broker funds and client funds are strictly separated
- SEBI circular compliance tracker for all new regulatory changes with implementation deadline tracking
- KYC annual refresh reminder dispatched 30 days ahead with automatic deactivation on non-compliance

FR9: Research and Investment Advisory System

Description: The system shall provide a research portal for analyst recommendations and investment strategy content.

Detailed Requirements:

- Research report creation by analysts with target company, target price, rating, and investment thesis
- Report approval workflow: analyst prepares, senior analyst reviews, compliance officer approves before publish
- Published reports visible to all active clients on the research portal with PDF download capability
- Model portfolio creation with defined allocation percentages per scrip and rebalancing triggers
- Target price tracking with automatic client alert when market price reaches the analyst's target
- Earnings calendar integration displays upcoming company results and consensus analyst estimates
- Sector performance heat map for visual comparison of relative sector returns over selected period
- Watchlist management: clients save up to 100 stocks with configurable price threshold alerts
- Research analytics: report view counts, target achievement rates, and recommendation accuracy tracking

- SEBI Research Analyst Regulation compliance: mandatory conflict of interest disclosures on all reports

FR10: Tax Reporting and Capital Gains Management System

Description: The system shall compute and present capital gains for income tax return preparation and filing assistance.

Detailed Requirements:

- LTCG versus STCG classification: equity held more than 12 months is LTCG; 12 months or less is STCG
- LTCG grandfathering computation for equity positions per the FY18 SEBI fair market value notification
- FIFO cost basis method applied consistently for all capital gains calculations across the portfolio
- Scrip-wise P&L statement showing buy cost, sell price, holding period, gain type, and net gain amount
- F&O trading profit and loss aggregated as business income for Income Tax Schedule BP purposes
- Dividend income tracking from all equity and mutual fund holdings for Schedule OS in ITR
- TDS reconciliation from Form 26AS for all dividend and interest income received in the year
- Tax harvesting identification: flags unrealised losses that can offset realised gains before March 31
- Advance tax liability estimate calculated from all realised gains at the current date in the FY
- ITR-compatible capital gains report in prescribed format for use by tax professionals and self-filers

FR11: Client Onboarding and Digital KYC Management System

Description: The system shall manage digital client onboarding with complete SEBI-compliant KYC verification.

Detailed Requirements:

- Digital account opening form captures personal details, financial profile, nominee, and bank account
- PAN verification via Income Tax Department API for name and date of birth cross-check
- DEMAT account verification via CDSL or NSDL API for linking an existing depository account
- CKYC registry check for clients who have completed KYC with any other SEBI-regulated entity
- Aadhaar OTP verification for current address confirmation and identity authentication
- Video KYC (IPV) with licensed KYC agent via WebRTC for final account activation
- Risk profiling questionnaire generates Conservative, Moderate, or Aggressive client classification
- Bank account penny-drop verification for payout account before first fund withdrawal is permitted
- Welcome email with trading ID, login credentials, and platform quick-start guide
- Annual KYC refresh reminder dispatched 30 days before anniversary date

FR12: Notification and Alert Management System

Description: The system shall deliver timely investment and account management notifications across all channels.

Detailed Requirements:

- Order execution confirmation via SMS and app push within 1 second of exchange acknowledgement
- Margin call alert at 80% utilisation via SMS, email, and in-app notification simultaneously
- Price alert delivered when a watchlisted stock reaches the client's configured target price
- Dividend credit notification when corporate action proceeds are posted to the client ledger
- Monthly portfolio statement emailed automatically on the 1st of each calendar month
- Research report publication alert to subscribed clients when a new report is approved and published
- Fund transfer confirmation via SMS and email with unique transaction reference number
- Circuit breaker notification to clients holding a stock when it enters upper or lower circuit
- KYC expiry reminder dispatched 30 days before the mandatory annual KYC refresh deadline

- Regulatory maintenance notification sent 24 hours before any scheduled system downtime

FR13: Mobile Trading and Remote Access System

Description: The system shall provide a secure mobile trading application for clients and dealers.

Detailed Requirements:

- Native Android and iOS trading application with biometric authentication on every app launch
- Real-time stock quotes, candlestick charts, and market depth (Level 2 order book) on mobile terminal
- Order placement with all order types including stop-loss, bracket, and cover orders on mobile
- Portfolio overview with live MTM P&L, current holdings, and sector allocation pie chart on mobile
- Fund transfer: NEFT and UPI integration for fund deposits and withdrawal requests from mobile
- Research portal with report download and watchlist management accessible on mobile application
- Configurable price alerts set from mobile with immediate push notification delivery on trigger
- Offline portfolio view shows last synchronised data when market data connection is unavailable
- Touch ID and Face ID as second factor authentication after initial password login on mobile
- Mobile app performance analytics track feature usage, session length, order placement, and crashes

System Administration and Security

FR14: Exception Handling and Error Management System

Description: The system shall implement comprehensive exception handling for stock brokerage and portfolio management operations.

Detailed Requirements:

- Name validation rejects non-alphabetic characters: 'Name must not contain special characters or numbers'
- Phone validation enforces 10 digits: 'Phone Number must be exactly 10 digits long'
- Duplicate Trade exception prevents creation of records with existing unique identifiers
- System exceptions handle database failures and external API timeouts with graceful degradation
- Custom exceptions in com.examly.springapp.exception: InvalidNameException, InvalidPhoneException, DuplicateTradeException, UnauthorisedAccessException
- Structured error logs capture exception type, user context, affected record, and timestamp
- User-facing messages are clear and actionable without exposing system internals
- Automatic retry with exponential back-off for transient external service failures
- Exception analytics track frequency by type, affected roles, and mean resolution time
- Integration error handling manages external API failures with logging and alert escalation

FR15: Data Security and Compliance System

Description: The system shall implement comprehensive data protection for stock brokerage and portfolio management data.

Detailed Requirements:

- AES-256 encryption at rest for all sensitive data fields in the database storage layer
- TLS 1.3 for all client-server communications with HSTS header enforcement
- Role-based API access control rejects every unauthorised request at the service layer
- Input sanitisation on all user inputs prevents SQL injection and cross-site scripting
- Security audit logs record every data access event with user identity and stated purpose
- Real-time intrusion detection monitors for anomalous access patterns and auto-blocks threats
- Data retention policies enforced with automated archival and documented deletion procedures

- Encrypted backup archives stored in geographically separate data centres
- Monthly automated vulnerability scans and annual professional penetration testing
- Compliance reporting generates audit packages for regulatory inspection on request

FR16: System Administration and Configuration Management

Description: The system shall provide comprehensive administrative tools for user lifecycle and compliance.

Detailed Requirements:

- User account lifecycle handles creation, role changes, suspension, and deactivation with audit trail
- System configuration allows parameter updates through admin portal without code redeployment
- Administrative dashboard provides real-time overview of users, activity, and system health metrics
- Bulk data import supports CSV-based mass record creation with validation and error reporting
- System health monitoring tracks API response times, error rates, and storage utilisation
- Automated daily encrypted backups with weekly restoration testing
- Audit log management provides searchable tamper-evident trail of all system changes
- Integration monitoring dashboard displays status of all external API connections
- Notification template management for all system-generated communications
- Compliance calendar tracks regulatory submission deadlines and licence renewals

FR17: Advanced Analytics and Predictive Intelligence System

Description: The system shall provide advanced analytics and predictive intelligence for stock brokerage and portfolio management decision support.

Detailed Requirements:

- Operational dashboard shows real-time stock brokerage and portfolio management KPIs and trend charts for management
- Custom report builder allows parametric reports with date range and dimension filters
- Data export in PDF, Excel, and CSV for regulatory and management reporting requirements
- ML-powered anomaly detection identifies unusual patterns requiring management attention
- Predictive modelling forecasts future demand and resource requirement trends
- Comparative benchmarking against industry or peer-group performance standards
- User behaviour analytics identify workflow bottlenecks and optimisation opportunities
- Automated weekly insight reports delivered to configured management stakeholders
- Risk scoring ranks records by composite risk factors for prioritised intervention
- Strategic planning tools generate evidence-based recommendations from analytics data

3.3 Performance Requirements

- Authentication: Login < 1 second for 95% of requests
- Record management: Create, read, update, delete < 2 seconds for 90% of requests
- Search: Results returned < 1.5 seconds for complex multi-field queries
- Validation: Combined client and server validation < 1 second for 95% of submissions
- Concurrent users: 500+ simultaneous users without performance degradation
- Reports: Standard reports generate in < 10 seconds for typical data volumes
- Notifications: End-to-end delivery within 5 seconds of trigger event

3.4 Logical Database Requirements

Core Tables:

- ClientAccounts Table: id, client_id, pan_number, demat_id, trading_status, available_margin, ledger_balance, risk_profile, kyc_status
- TradeOrders Table: id, client_id, symbol, order_type, transaction_type, quantity, price, status, placed_at, executed_at

- Holdings Table: id, client_id, symbol, quantity, avg_cost, current_price, unrealised_pnl, last_updated
- Ledger Table: id, client_id, transaction_type, debit, credit, balance, narration, transaction_date

3.5 Design Constraints

- Spring Security + JWT on all protected endpoints
- React with protected routes and mobile-responsive design
- JPA transactions for concurrent data integrity
- All changes logged with user identity and timestamp
- Name: alphabetic only; Phone: exactly 10 digits
- RBAC enforced at both React route and Spring Boot API level

3.6 Software System Attributes

3.6.1 Reliability

- 99.9% uptime with planned maintenance communicated 48 hours ahead
- Graceful non-sensitive error messages for all system failures
- Multi-layer data integrity validation at client, API, service, and database levels
- Automatic failover for critical stock brokerage and portfolio operations within 60 seconds
- Daily encrypted backups with quarterly restoration testing

3.6.2 Availability

- 24/7 availability with maintenance outside peak operational hours
- Load balancing across minimum two application server instances
- Geographic data replication to secondary data centre for disaster recovery
- Real-time health monitoring with automated on-call escalation
- Horizontal auto-scaling for peak load periods

3.6.3 Security

- Authentication: JWT HS256 with role-specific session duration configuration
- Authorisation: Role-based access control at API service layer for every endpoint
- Data Protection: AES-256 at rest and TLS 1.3 in transit for all sensitive data
- Input Validation: Server-side sanitisation against injection and XSS attacks
- Audit Trail: Immutable logging of all data access and modification events
- Compliance: Regulatory standards verified and documented at every major release

3.6.4 Maintainability

- Layered architecture: Controller, Service, Repository, Domain Model layers
- Minimum 85% unit and integration test coverage measured in CI pipeline
- Swagger/OpenAPI 3.0 documentation with examples for every endpoint
- Git feature branch workflow with mandatory peer code review
- Spring Boot profiles with encrypted secret management

3.6.5 Portability

- Docker + Docker Compose for consistent local and production deployments
- Spring Data JPA: MySQL 8.0+ and PostgreSQL 14+ without code changes
- Cloud manifests: AWS ECS, Azure AKS, GCP GKE
- Spring Boot profiles: dev, test, staging, production
- Cross-browser: Chrome 100+, Firefox 100+, Edge 100+, Safari 15+

3.7 Other Requirements

- Testing: Backend 85% JUnit 5 and Mockito; Frontend 75% React Testing Library
- Security Testing: Quarterly OWASP ZAP + annual professional penetration testing
- Performance Testing: Monthly JMeter load tests before each major release

- Documentation: API reference, deployment runbook, user manuals each release
- Compliance: Finance regulatory standards verified at every major release
- Accessibility: WCAG 2.1 Level AA for all user-facing interfaces

4. Appendices

Appendix A: Role-Based Permission Matrix

Functionality	Guest	Client	Dealer	Research Analyst	Compliance Officer	Risk Manager	Admin
View Market Data	✓	✓	✓	✓	✓	✓	✓
Place Trade Order	-	✓	✓	-	-	-	✓
View Own Portfolio	-	✓	✓	L	L	L	✓
View All Client Portfolios	-	-	✓	-	L	✓	✓
Publish Research Reports	-	-	-	✓	-	-	✓
Monitor Client Margins	-	-	L	-	-	✓	✓
Generate Regulatory Reports	-	-	-	-	✓	-	✓
Configure Risk Limits	-	-	-	-	-	✓	✓
Manage Users	-	-	-	-	-	-	✓
View Audit Logs	-	-	-	-	✓	-	✓

Appendix B: Database Schema Details

ClientAccounts Table:

- id (PRIMARY KEY, AUTO_INCREMENT)
- client_id (FOREIGN KEY → Users.id, NOT NULL)
- pan_number (VARCHAR(10), UNIQUE, NOT NULL)
- demat_id (VARCHAR(20), UNIQUE, NOT NULL)
- trading_status (ENUM: ACTIVE, SUSPENDED, CLOSED, NOT NULL)
- available_margin (DECIMAL(15,2) DEFAULT 0)
- ledger_balance (DECIMAL(15,2) DEFAULT 0)
- risk_profile (ENUM: CONSERVATIVE, MODERATE, AGGRESSIVE)
- kyc_status (ENUM: PENDING, VERIFIED, NOT NULL)

TradeOrders Table:

- id (PRIMARY KEY, AUTO_INCREMENT)
- client_id (FOREIGN KEY → Users.id, NOT NULL)
- symbol (VARCHAR(20), NOT NULL)
- order_type (ENUM: MARKET, LIMIT, STOP_LOSS, BRACKET)
- transaction_type (ENUM: BUY, SELL, NOT NULL)
- quantity (INT, NOT NULL)
- price (DECIMAL(10,2))
- status (ENUM: PENDING, EXECUTED, CANCELLED, REJECTED)

- placed_at (TIMESTAMP, NOT NULL)
- executed_at (TIMESTAMP)

Holdings Table:

- id (PRIMARY KEY, AUTO_INCREMENT)
- client_id (FOREIGN KEY → Users.id, NOT NULL)
- symbol (VARCHAR(20), NOT NULL)
- quantity (INT, NOT NULL)
- avg_cost (DECIMAL(10,4), NOT NULL)
- current_price (DECIMAL(10,4))
- unrealised_pnl (DECIMAL(12,2))
- last_updated (TIMESTAMP)

Ledger Table:

- id (PRIMARY KEY, AUTO_INCREMENT)
- client_id (FOREIGN KEY → Users.id, NOT NULL)
- transaction_type (VARCHAR(50), NOT NULL)
- debit (DECIMAL(12,2))
- credit (DECIMAL(12,2))
- balance (DECIMAL(12,2), NOT NULL)
- narration (VARCHAR(200))
- transaction_date (TIMESTAMP, NOT NULL)

Appendix C: Validation Rules Summary

Field	Validation Rules	Error Messages
Name	Required, alphabetic characters and spaces only, 2–100 characters	"Name is required" / "Name must not contain numbers or special characters"
Phone Number	Required, exactly 10 consecutive numeric digits	"Phone Number is required" / "Phone Number must be exactly 10 digits long"
Email	Required, valid email format, unique per account	"Please enter a valid email address" / "This email is already registered"
Password	Minimum 8 characters with uppercase, lowercase, digit, and special character	"Password must meet security requirements"
Trade Quantity	Positive integer within applicable lot size	"Quantity must be a positive integer within the lot size"
Limit Price	Positive decimal number	"Limit price must be a positive number"
Fund Transfer Amount	Positive decimal within available ledger balance	"Amount must be positive and within available balance"

Appendix D: Security Implementation Details

JWT Configuration:

- Token Expiration: 8 hours for regular users, 12 hours for staff, 24 hours for admins

- Refresh Token: 30-day validity with automatic rotation on each use
- Algorithm: HS256 with minimum 256-bit secret key stored as environment variable
- Secret Key: Never hardcoded in source code — loaded from secure environment config
- Token Blacklisting: Implemented for logout, password change, and security breaches
- Token Claims: Include userId, role, email, issuedAt, and expiration timestamp

Password Policy:

- Minimum 8 characters with uppercase, lowercase, digit, and special character
- Cannot match last 5 passwords; expires 90 days staff / 180 days users
- Lockout 15 minutes after 5 consecutive failed login attempts
- Secure password reset via email OTP with 15-minute validity window
- Password strength indicator displayed in real time during registration

Data Security:

- Sensitive stock brokerage and portfolio data encrypted at rest using AES-256
- All communications use HTTPS with TLS 1.3 and HSTS headers
- Server-side input sanitisation preventing SQL injection and XSS
- Role-scoped data access controls enforced at service layer
- Tamper-evident encrypted audit logs for all security events

Rate Limiting:

- Login: maximum 5 attempts per 15-minute window per IP address
- Registration: maximum 3 new accounts per day per IP
- Search: maximum 1,000 queries per hour per authenticated user
- Data writes: maximum 100 operations per hour per user
- Password reset: maximum 3 emails per hour per address

Appendix E: Exception Handling Classes

The following custom exception classes shall be implemented in `com.examly.springapp.exception`:

InvalidNameException

Thrown when a stock brokerage and portfolio name field contains non-alphabetic characters or digits.

- Extends: `RuntimeException`
- Trigger: Non-alphabetic characters in name field
- HTTP Status: 400 Bad Request

InvalidPhoneException

Thrown when a phone number is not exactly 10 consecutive digits.

- Extends: `RuntimeException`
- Trigger: Phone not exactly 10 digits
- HTTP Status: 400 Bad Request

DuplicateTradeException

Thrown when a stock brokerage and portfolio record with the same unique identifier already exists.

- Extends: `RuntimeException`
- Trigger: Duplicate unique identifier on create
- HTTP Status: 409 Conflict

UnauthorisedAccessException

Thrown when a user accesses a resource outside their assigned role.

- Extends: RuntimeException
- Trigger: API call beyond user role permissions
- HTTP Status: 403 Forbidden

ResourceNotFoundException

Thrown when a stock brokerage and portfolio record cannot be found by the provided ID.

- Extends: RuntimeException
- Trigger: GET/PUT/DELETE with non-existent ID
- HTTP Status: 404 Not Found

Appendix F: Business Rules and Constraints

Trading Rules:

- Order execution requires sufficient margin validated in real time before the order is accepted
- All executed trade orders are immutable — cancellation permitted only before execution
- Stop-loss orders auto-execute at the configured trigger price during market hours
- Position limits enforce SEBI client-level concentration norms per scrip and sector
- Post-market orders are queued and execute at the next trading session open price

Settlement Rules:

- T+1 equity trade settlement as per SEBI mandate: proceeds credited next working day
- Brokerage and all statutory charges deducted from client ledger on trade execution
- Fund settlement processed via NEFT or UPI within T+1 after trade confirmation
- Corporate actions (dividends, bonuses, splits) processed within 5 working days of record date
- Short delivery triggers exchange auction settlement as per NSE and BSE guidelines

Compliance Rules:

- PAN and verified DEMAT account mandatory before any trade order is accepted and processed
- SEBI daily activity report submitted to exchange by end of every trading day
- Client funds must be maintained in a segregated account separate from broker own funds at all times
- Margin calls must be met within 1 full trading session or at-risk positions are squared off
- Annual KYC refresh mandatory for all active trading accounts as per SEBI circular

Appendix G: Future Enhancement Roadmap

Phase 2 Features:

- Native iOS and Android apps with offline stock brokerage and portfolio capability
- Advanced analytics with machine learning trend forecasting
- Third-party platform integration via REST and webhook APIs
- Automated recommendation algorithms reducing manual work
- Multi-language support for diverse user populations

Phase 3 Features:

- AI for automated pre-screening and intelligent workflow routing
- Predictive analytics for outcome forecasting
- Blockchain-based immutable record management
- Real-time collaborative workspaces with co-editing

Phase 4 Features:

- Continuous ML retraining pipelines from production data
- Self-service BI with drag-and-drop report builder
- AI-assisted data extraction reducing manual entry by 80%
- Open partner API marketplace for third-party extensions

Appendix H: API Endpoint Summary

Trading Operations:

- POST /api/orders — Place a new trade order
- GET /api/orders/{id} — Get order details
- DELETE /api/orders/{id} — Cancel a pending order
- GET /api/orders/client/{id} — Client full order history
- GET /api/orders/book — Today's complete order book

Portfolio and Ledger:

- GET /api/portfolio/{clientId} — Holdings with real-time MTM P&L
- GET /api/portfolio/performance/{clientId} — Return metrics and analytics
- GET /api/ledger/{clientId} — Ledger statement with date filters
- POST /api/funds/transfer — Initiate fund in or fund out request
- GET /api/portfolio/tax/{clientId}/year/{year} — Annual tax P&L report

Market Data and Research:

- GET /api/market/quote/{symbol} — Real-time stock quote
- GET /api/market/chart/{symbol} — OHLCV historical chart data
- GET /api/research — Published research reports list
- GET /api/margin/{clientId} — Current margin utilisation
- GET /api/risk/alerts — Active margin and risk alerts for risk manager

User Management:

- POST /api/auth/register — Register new user account
- POST /api/auth/login — Authenticate and receive JWT token
- POST /api/auth/logout — Invalidate current JWT token
- GET /api/users/profile — Get current user profile

Appendix I: Component and Navigation Structure

React Components:

App.jsx:

- React Router DOM v6 with nested and protected route support
- Routes: "/" (Home), "/login" (Login), "/register" (Register), "/dashboard" (Dashboard), "/trade" (TradingTerminal), "/portfolio" (Portfolio)

- NavBar and Footer in root layout for all pages
- ProtectedRoute redirects unauthenticated users to login
- Global ErrorBoundary for unhandled runtime exceptions

NavBar.jsx:

- Application title: "Stock Brokerage and Portfolio Management System"
- Navigation links: "Home" (/), "Login" (/login), "Register" (/register), "Dashboard" (/dashboard), "Trade" (/trade), "Portfolio" (/portfolio)
- Logged-in user name and role from JWT token in AuthContext
- Logout clears JWT from localStorage and redirects to login
- CSS classes: nav-link, nav-active, nav-brand

Login.jsx:

- Email/username and password with show/hide toggle
- Submit "Login" → POST /api/auth/login
- JWT stored in localStorage; role saved in AuthContext on success
- Redirects to role-specific dashboard from JWT role claim
- Error: "Invalid credentials. Please check your email and password."

Register.jsx:

- Multi-step form: name, email, phone, role, domain-specific fields
- Name: alphabetic + spaces only; Phone: exactly 10 digits
- Email uniqueness verified via debounced API call
- Submit "Register" → POST /api/auth/register
- Success: "Registration successful! Please verify your email."

Dashboard.jsx:

- Role-aware cards, panels, and data tables per authenticated role
- KPI summary cards for most relevant metrics per role
- Recent activity feed with links to detail pages
- Quick-action buttons for three most-used operations
- Responsive CSS grid: single-column mobile, three-column desktop

Footer.jsx:

- "© 2024 Stock Brokerage and Portfolio Management System. All rights reserved."
- Rendered via App.jsx layout — not imported individually
- Links to privacy policy, terms of service, and support

Document Version Control

Version	Changes	Author
1.0	Initial SRS document creation for Stock Brokerage and Portfolio Management System	Development Team